HEINZ NIXDORF INSTITUT
UNIVERSITÄT PADERBORN

# Knowledge Graphs: Construction, Reasoning, and Retrieval-Augmented Generation

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Tutor: Asep Fajar Firmansyah

DICE
Data Science Group
Paderborn University

**Project Group - WiSe 2026**
July 27, 2026

# About DICE



What does the DICE group do?

- Data Integration
- Data Storage and Querying
- Interpretable Data Processing
- Machine Learning
- Natural Language Processing

For more information: https://dice-research.org/groups/

Section 1

# PG Organization

# PG Structure and Deliverables



**How is this project group structured?**

- **Semesters:** WiSe 26/27 and SoSE 27
- **Registration:** You need to register your self before 31.10.2026.
- **During Semesters:**
  - Regular Group Meetings – (Usually) Every Two Weeks
  - Weekly Report
- **Semester-end Requirements:**
  - Individual Report, Group Report, and Group Presentation
  - Grading and Feedback
- **Final Submission:** Source Code and Documentation

# Expectations



**What do we expect from you?**

→ Be a valuable member of your team!

- Manage your project
- Write code(!) and commit it
- Communicate with team members
- Support each other where possible
- …

→ We won't keep students if they are not participating in their team

# PG Schedule



- **Kickoff Meeting:** 15.10.2026, FU.519
- **Regular Meetings:** Data DBD, FU.519
- **Registration:** Studienleistung and Register for Exam before the deadline—or leave the PG!
- **End WiSe 26/27:**
  - Individual Report and Group Presentation
  - Grading and Feedback
- **Between WiSe 26/27 and SoSe 27:** Vacations
- **Start SoSe 27:** Resume PG Work
- **Mid-July 2027:** End of the Implementation Phase
- **End SoSe 26:**
  - Group Report and Group Presentation
  - Final Grading

# Unforeseen Events



- **Vacations:**
  - Should be avoided during lecture time!
  - In very urgent cases, contact your supervisor
- **Project Issues:**
  - Problems within the team
  - Hardware issues
  - Personal issues

  $\rightarrow$ Get in touch with your supervisor
- **Illness:**
  - Get a Doctor's Certificate!
  - If you can foresee it, inform your teammates

Section 2

# PG Objectives Recap

# Knowledge Graphs

## Introduction



A Knowledge Graph (KG) represents entities as nodes and their relationships as edges.

- Knowledge Graphs Everywhere



- Many open knowledge graphs




- Powering AI




- Applied across diverse domains

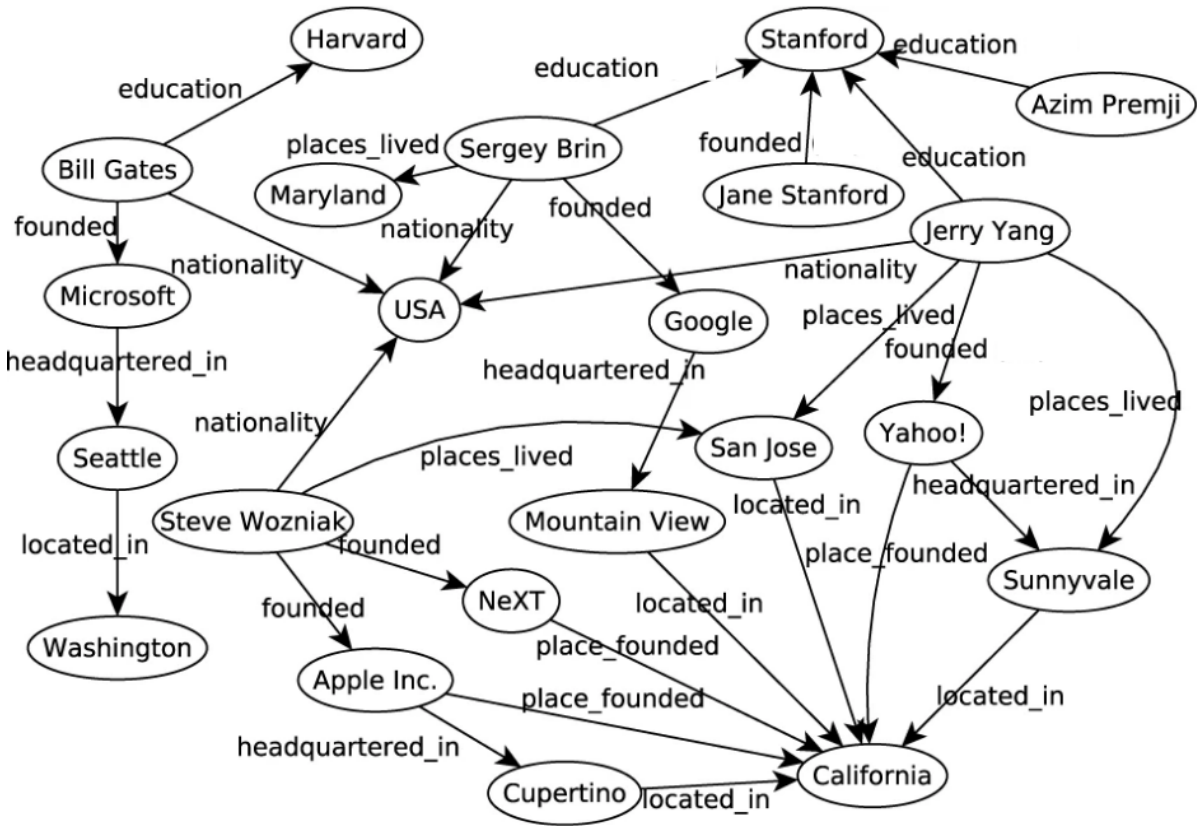


# Knowledge Graphs

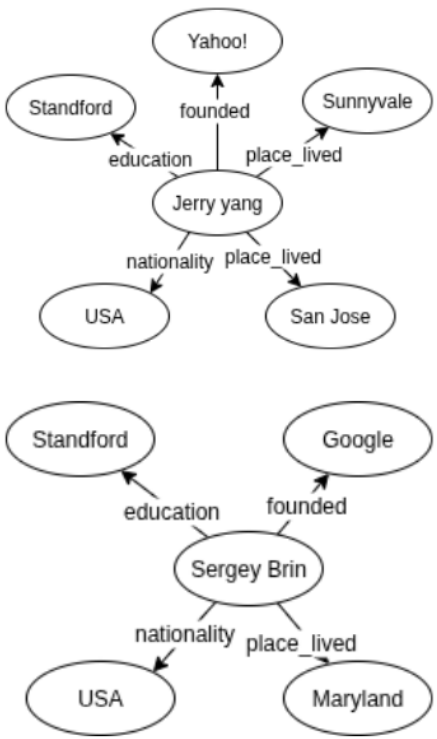
## Structure and Representation



### A knowledge Graph




### Entity Description



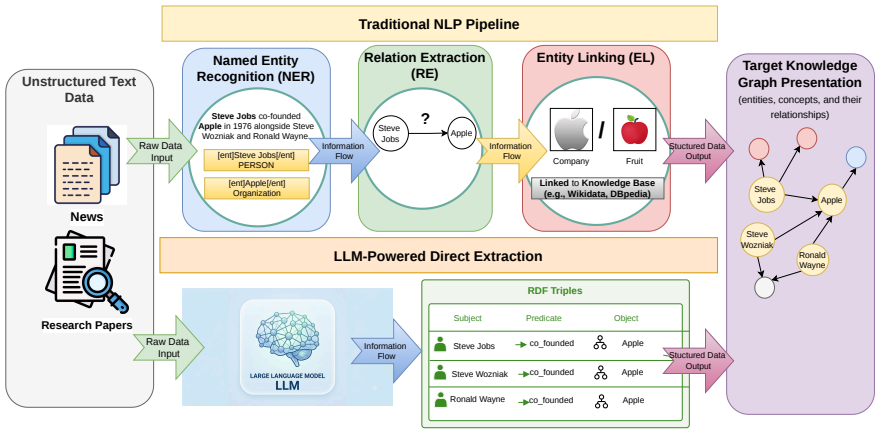
# Knowledge Graphs: Construction



## Why we need to construct the knowledge graph?

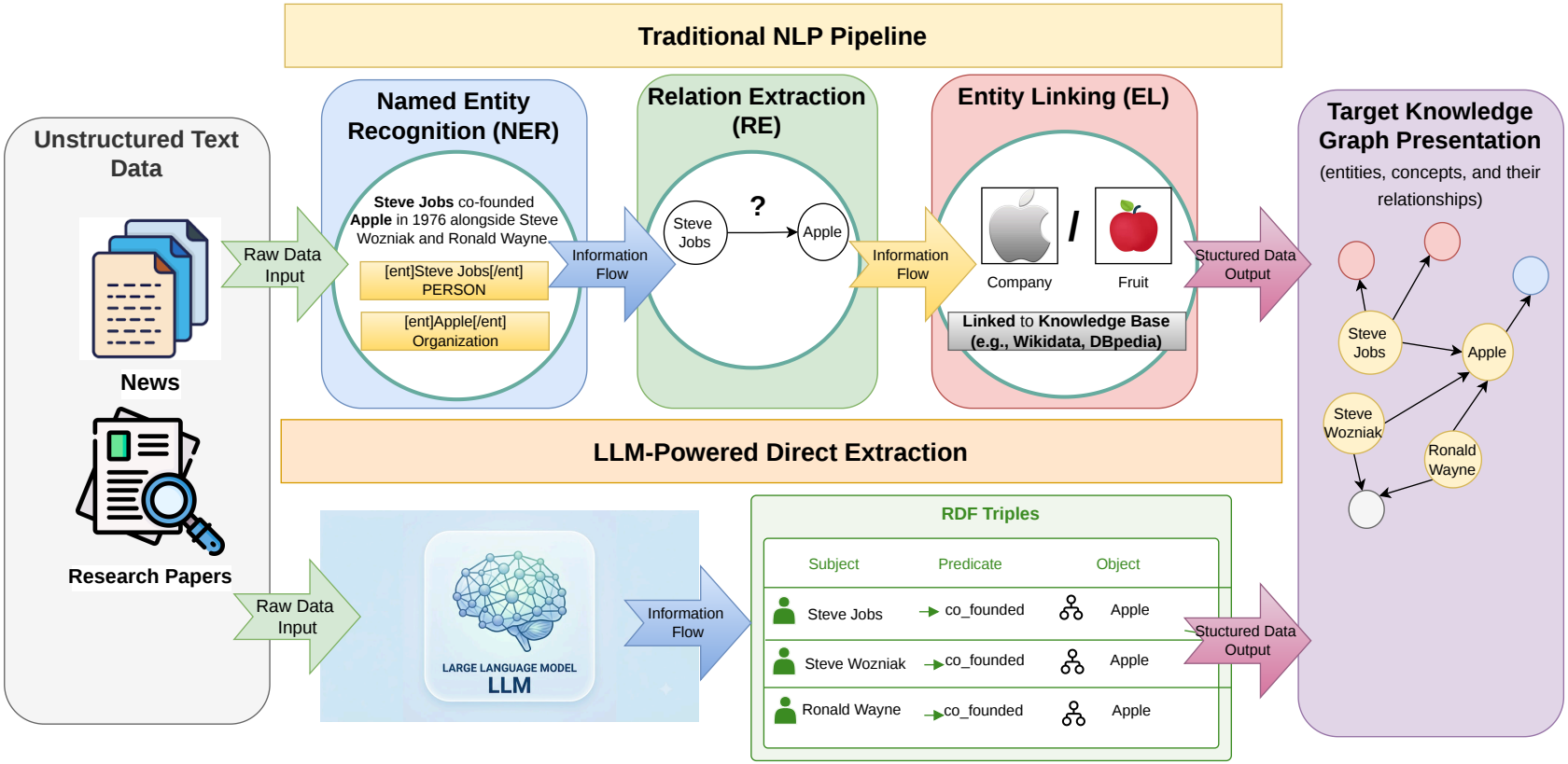
### Motivation

- **Problem**: Vast amounts of valuable domain knowledge remain trapped in unstructured raw text.
- **Solution**: Extract structured RDF triples using NLP pipelines (NER, RE, EL) or direct LLMs.
- **Goal**: Build machine-readable, interconnected knowledge graphs from unstructured data sources.



# Knowledge Graphs: Construction

## Approaches



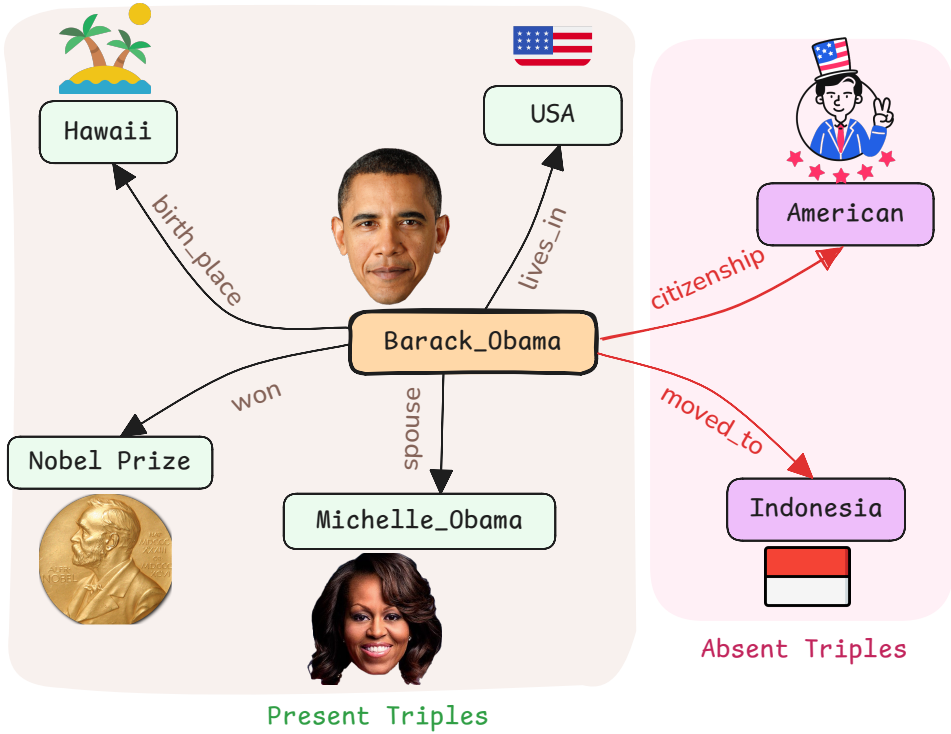

Tiwari, B., Oliviera, I., Firmansyah, A.F., Zahera, H.M., HOPFGARTNER, F., Ngomo, AC.N. (2026). Ontology-Aware Prompting for Knowledge Graph Construction from Text. Accepted at SEMANTICS 2026.

# Knowledge Graphs: Reasoning



## Motivation

- **Problem**: Incompleteness and missing links in Knowledge Graphs limit downstream tasks.
- **Solution**: Apply logical inference and machine learning (GNNs/Embeddings) for link prediction.
- **Goal**: Infer unstated facts, discover hidden patterns, and complete missing graph relationships.



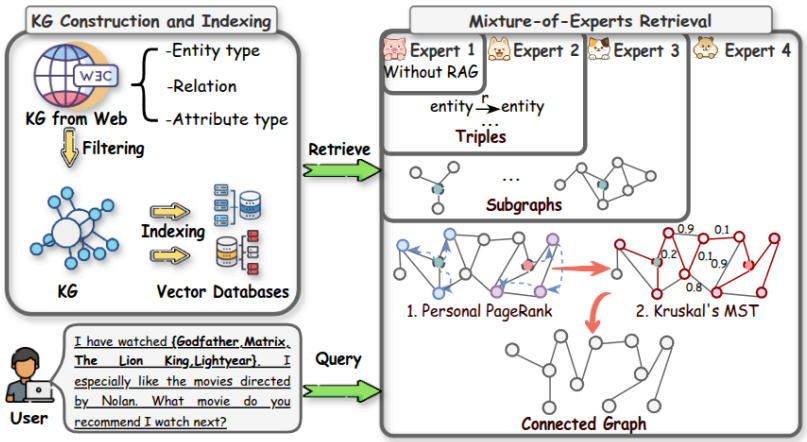

*Firmansyah, A.F., Zahera, H.M., Sherif, M.A., Moussallem, D., Ngomo, AC.N. (2025). ANTS: Abstractive Entity Summarization in Knowledge Graphs. The Semantic Web – ESWC 2025. LNCS, vol. 15718, Springer, Cham.*

# KG Retrieval-Augmented Generation



## Motivation

- **Problem**: LLMs suffer from hallucinations, lack domain-specific knowledge, and cannot cite explicit sources.
- **Solution**: Retrieve structured, factual subgraphs from Knowledge Graphs to ground LLM prompts in verifiable context.
- **Goal**: Generate accurate, explainable, and hallucination-free answers enriched with structured semantic knowledge.




Source: https://arxiv.org/pdf/2605.28175

# Project Objectives



Addressing the core challenges of Knowledge Graph construction, reasoning, and RAG across the complete research lifecycle.

- **Enhance performance:** Improve extraction accuracy, logical inference capability, and factual precision in LLM generation.
- **Explore Advanced Techniques:** Leverage LLMs for automated RDF extraction, Deep Learning for KG reasoning, and KG-RAG architectures.
- **Discover Novel Approaches:** Innovate at the intersection of structured knowledge bases and Generative AI.

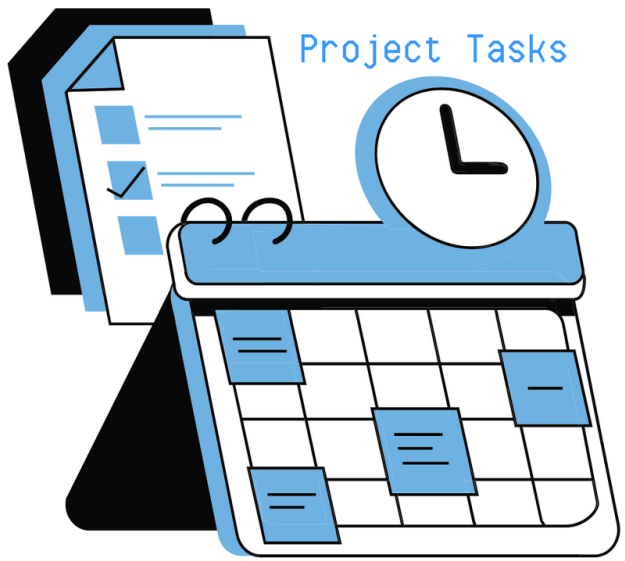


# Project Tasks



What core tasks will the project group undertake across the research lifecycle?

- **Literature Study:** Survey state-of-the-art methods.
- **Baseline Reproduction:** Implement and test existing baselines on benchmark datasets.
- **Gap Analysis:** Identify limitations in current systems.
- **Method Development and Evaluation:** Design, implement, and evaluate improved pipelines for automated construction, reasoning, or KG-RAG.
- **Graph Visualization**



# We Offer



- Training Computer Resources
- Follow-up Thesis Opportunities
- Publication Support



Section 3

# Goals & Grading

# PG Goals



## Current Semester

## What are the goals for the current semester?

1. Investigate KG construction, reasoning, and RAG approaches.
2. Run experiments and evaluate each approach.
3. Provide in-depth analysis on what works better and what doesn't.

# Grading



**How will the students be graded?**

- **Achieved Goals:**
  - **Group:** What goals did the group achieve?
  - **Individual:** How much did you contribute towards this goal?
- Reports and Presentations (Group, Individual)
- **Soft Skills (Group, Individual):**
  - Commitment and engagement
  - Collaborative and learning skills
  - Motivation and volitional strategies
  - Scientific writing and self-management

Section 4

# Ice Breaker

# First Task



*"Talk is cheap. Show me the code."* – Linus Torvalds

1. Choose a paper based on your interests.
2. Read, learn, and investigate the content
3. Prepare for the colloquium

→ **Time Limit:** 2 Week

# Next Steps



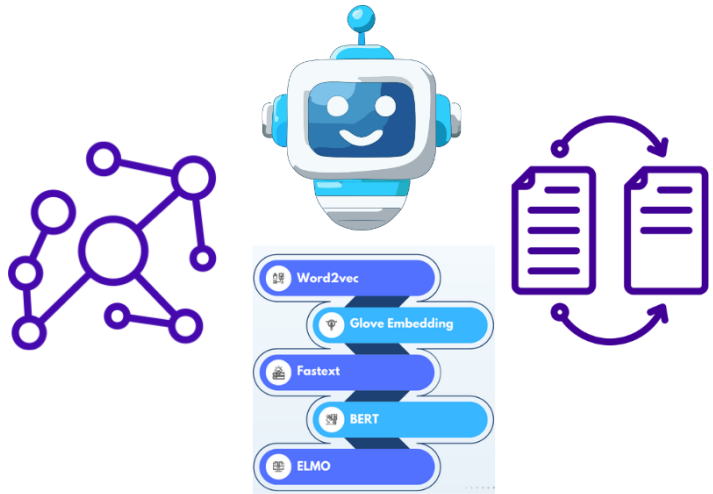
*"A place for everything, everything in its place"* – Benjamin Franklin

→ Organize Yourselves

- Team Structure: Flat, Hierarchical, Holacratic, …
- Communication: UPB Matrix
- Code Repository: https://github.com/dice-group/KGCR-RAG-2026

→ Try to use open-source tools wherever possible

# That's all Folks!




Thank you!

dice-research.org/teaching/KGCR-RAG-2026

**Have questions?**

Email: asep.fajar.firmansyah@uni-paderborn.de
Matrix: @asepff:chat.dice-research.org

*All source images are from Google Search and Canva.*

# Project Group Use Case



